

Week 17 – Mock Contest 1

■ Contest Setup (1.5 hrs, 3 problems: J1–J3)

- No internet or AI tools; only standard Python.
- Aim for correctness first, then minor optimizations.
- Manage time: ~20–25 min per problem, buffer at end.

Problem A (J1) – Score Calculator

Read N pairs , output each name's total and the winner.

Solution:

```
n = int(input())
tot = {}
for _ in range(n):
    name, s = input().split()
    s = int(s)
    tot[name] = tot.get(name, 0) + s
# print totals
for k in sorted(tot):
    print(k, tot[k])
# winner
print("Winner:", max(tot, key=tot.get))
```

Problem B (J2) – Mirror String

Given a string, print it reversed and also with each letter's alphabet index (a=1).

Solution:

```
s = input().strip().lower()
rev = s[::-1]
print(rev)
indices = []
for ch in s:
    if ch.isalpha():
        indices.append(str(ord(ch) - 96))
print(" ".join(indices))
```

Problem C (J3) – Grid Walk

Given a grid with obstacles (#), start (0,0). Count reachable cells using BFS (4 directions).

Solution:

```
from collections import deque
R, C = map(int, input().split())
grid = [list(input().strip()) for _ in range(R)]
q = deque([(0,0)])
seen = {(0,0)}
dirs = [(-1,0),(1,0),(0,-1),(0,1)]
while q:
    r,c = q.popleft()
    for dr,dc in dirs:
        nr, nc = r+dr, c+dc
        if 0 <= nr < R and 0 <= nc < C and grid[nr][nc] != '#' and (nr,nc) not in seen:
            seen.add((nr,nc)); q.append((nr,nc))
print(len(seen))
```

■ Debrief & Time Management

- Start with I/O reading, small test cases, then implement core logic.
- If stuck >7 minutes, switch to next problem and return later.
- Validate on edge cases: empty, single-element, boundaries, ties.

■ Homework

- Re-solve Problem B using dictionary mapping for $O(1)$ lookups.
- Turn Problem C into a function `reachable(grid)` and write 3 unit-like tests.